Table 12.1 Composition of the universe

Constituent	% of Universe mass
Dark energy	70
Dark matter	25
Free H and He	4
Stars	0.5
Neutrinos	0.3
Elements heavier than He	0.03

12.1. OBSERVATIONS FROM SPACE PROBE EARTH

12.1.1. Our place in the universe

The stars we directly can see with our eyes all belong to our Galaxy, the *Milky Way*, which is a barred spiral galaxy, about 30 kpc (kpc = kiloparsec) across and about 1 kpc thick, See Fig.12.1. The parsec (or parallax second) is the common astronomical unit of distance; 1 parsec is 3.1×10^{16} m, or 3.26 light years, ly, which corresponds to a one second change in direction angle when the earth moves between two opposite positions around the sun. Thus light travels across our galaxy in about 100 000 y. The Milky Way contains some 200×10^9 stars, interstellar dust and gas ($\sim$200 pc thick) which spreads out to a diameter of about 50 kpc (hot gas atoms and single stars, the *halo*). It has two long spiral arms and a short one beginning from the ends of a central bar. Our sun is located in a spur from the outer edge of the short spiral arm, about 8.5 kpc from the galactic center as shown in Figures 12.1 and 12.2. The dust limits the sight at visual wave-lengths towards the center to only a few kpc; without this dust the Galaxy center would shine equally bright as our sun. Infrared- and radio-telescopes have made it possible to penetrate much of the dust and show stars circling very rapidly around the center. The stars in the bar seems at the same time to rotate like a cylinder. The other stars in our Galaxy move tangentially around its center with angular velocities also increasing closer to the center. Both observations indicate the existence of an extremely heavy central object, called Sagittarius (Sgr) A^*.

The Milky Way belongs to the *Local Group*, a cluster of some 20 galaxies which include, i.a. the Large Magellanic Cloud, our nearest galaxy, 50 kpc away and the Andromeda galaxy, 650 kpc away. The Local Group is part of the larger *Virgo super-cluster*. The universe contains some 10^{10} galaxies, but galaxies, stars and gases only fill a fraction of space, <5%, the rest appears void of matter. On a grand scale it looks like the galaxies and empty space form a bubbly structure, see Figure 12.3.

In the 1930s Hubble discovered that galaxies on the whole are equally distributed in all directions of space as observed from the earth. Thus space – on a large scale – seem to be isotropic. This idea of uniformity of the universe is called *the cosmological principle*.

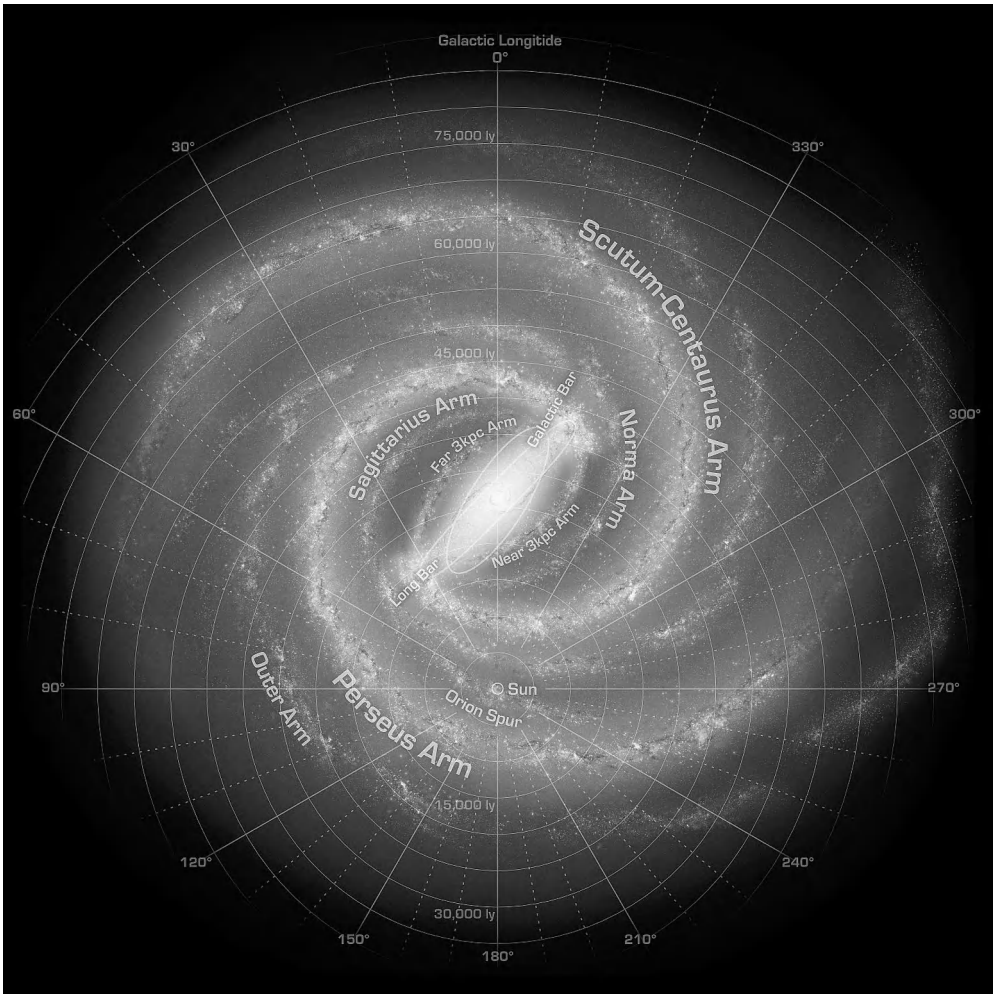


Figure 12.1 Sketch of our galaxy, the Milky Way, as seen from above (NASA).

This information has been deduced from *celestial mechanics* (movements of bodies according to Newton's fundamental laws) and from spectroscopic analysis of light and other kinds of radiation. It has been found that the mass of our sun is 1.99×10^{30} kg ($= 1$ solar mass, $\mathbf{M}_\odot$). The mass of the Milky Way is $>2 \times 10^{11}\ \mathbf{M}_\odot$. About 10% of the mass is interstellar gas, and 0.1% is dust (typically particles with diameter $0.01-0.1$ μm). The interstellar gas density varies considerably in our galaxy; in our part of space it varies from about 10^9 (in dark clouds) to 10^5 atoms m^{-3} (on the average ~ 1 atom cm^{-3}). Though it contains mainly H and He, also large rather complex molecules containing H, C (up to C_{15} molecules), N and O (including amino acids) have been discovered.

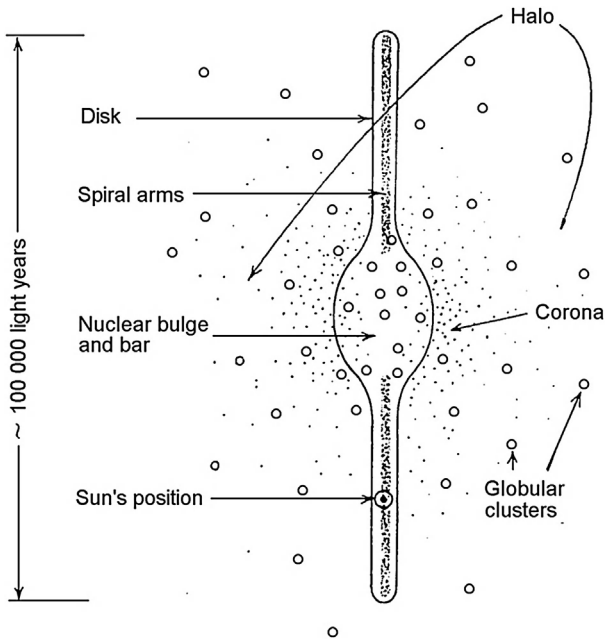


Figure 12.2 Sketch of the Milky Way as seen from the side (edge) and showing the halo.

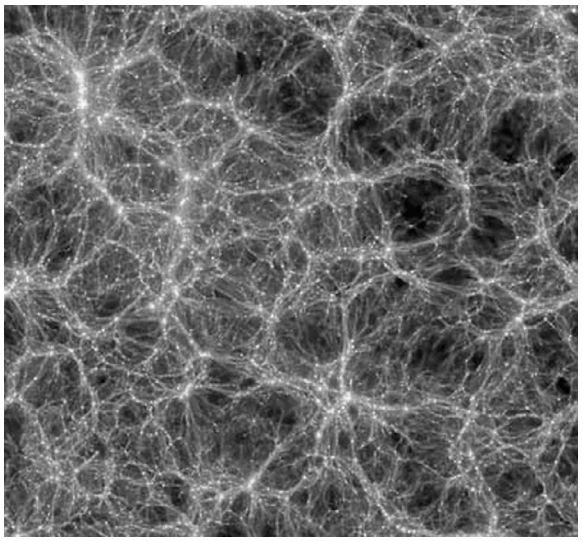


Figure 12.3 Sketch of the general structure of our universe showing its bubbly structure. Each object plotted is a galaxy.

12.1.2. Dark matter

Astronomical models of the universe indicate that it will expand forever if the observed galaxies alone should account for the total mass of the universe. Moreover, the visible mass within galaxies also seems to be too small to hold them together. Based on the average density of the observable universe, 90% of the mass needed for a slowing down and ultimately contracting universe is missing. Most cosmologists believe that the mass of the observed galaxies is $\leq$5% of the mass of the universe, the main part consisting of "*dark matter*"; this includes interstellar and intergalactic matter, brown dwarfs, *neutron stars*, "*black holes*" and other little known sources of radiation, like quasars, whose masses are unknown. Measurements of the cosmic background electromagnetic radiation indicate that the universe now is infinite and flat.

From mechanics and Newton's gravitational law one can calculate the velocity needed for a body, m_x, to escape the gravitational pull of a larger mass, m, where $m_x \ll m$. For example, if m is the earth's mass (5.94×10^{24} kg), a rocket (mass m_x) must have a velocity of about 11 km/s to escape from the earth's surface (the *escape velocity*, v_e). Conversely, for a given velocity, v_e, one can calculate the mass and size of the large body needed to hamper such an escape. A body with one solar mass, $\mathbf{M}_\odot$, but a radius of only 3 km, requires an escape velocity $>3 \times 10^{10}$ m/s. Thus not even light will escape such a body, which therefore is termed a *black hole*. Though we cannot observe the black holes directly, some secondary effects can be observed.

Astronomical observations of star movements support the existence of black holes. For example, from movements of stars close to our galactic center, it is believed that a black hole is located at SgrA* in the center of the Milky Way, with a mass $>3 \times 10^6\ \mathbf{M}_\odot$. The radius[1] of such a hole would be of the same size as that of our sun. The density of matter in the hole would be several million times the density of our sun (average value for the sun is about 1400 kg/m^3). Obviously matter cannot be in the same atomic state (i.e. nuclei surrounded by electrons) as we know on earth. Instead we must assume that the electron shells are partly crushed; we refer to this as *degenerate matter*, because the electron quantum rules, Tables 6.1–6.2, cannot be upheld. For completely crushed atoms, matter will mainly consist of compact nuclei. For example, for ^{40}Ca the average nuclear density is $\sim 7.3 \times 10^{16}$ kg/m^3 ($r \sim 6$ fm, cf. Fig. 4.4).

Even if black holes are given a considerable portion of the missing mass, this will not be enough. However, a recent discovery may perhaps provide the "needed" mass: Detailed analysis of the variation in luminosity (a factor of about 2.5) for some 10 million double stars in the Large Magellanic Cloud, gives support for the existence of nearby "*gravitational microlenses*", which are believed to be unborn stars (so-called *brown dwarfs*, §12.3) of sizes $\sim 10\ \mathbf{M}_\odot$. When such a dark object passes the line of sight to a distant star

[1] The radius at which light can no longer escape from the central gravity field.

it acts as a focusing lens for the light, thereby temporarily increasing that star's apparent luminosity. As these gravitational microlenses seem to be especially abundant in the halo of our galaxy (and presumably in halos of other galaxies), they — together with neutron stars and black holes — could correspond to a part of the dark-matter needed to hold galaxies and clusters of galaxies together by gravity.

12.1.3. Dark energy

Measurements of the distances and red-shifts for many distant galaxies, using type 1a supernovas as standard candles, indicate that the expansion of the universe is not decreasing as expected but expanding with increasing velocity. In practice this *red shift* is a Doppler effect shifting to longer wave lengths as a light source moves away from the observer, see eqn 12.2. The opposite is called *blue shift* and is the corresponding shift to shorter wave lengths as the source is moving close to the observer. Measurements and calculations indicate the there was a shift from retardation to expansion about 7.9×10^9 years ago.

The accelerating expansion of remote objects away from us is thought to be due to a weak repulsive force that penetrates the whole universe. Nothing is known about this force, but when it acts accelerating upon the universe an enormous energy is needed to overcome gravity and increase speed — the so called dark energy. When converted to mass by the Einstein equation it corresponds to about 70% of the total mass of the universe.

12.1.4. Light, energy flux and the hubble law

Spectral analysis of the light received from astronomical objects have provided us with information of their (surface) *temperature* (from their continuous spectrum) and outer *chemical composition* (from identification of spectral line frequencies), while bolometric measurements have given their *luminosity* (energy flux density, F Wm^{-2}). In 1911 Hertzsprung and Russell discovered that if the luminosity and colour (or temperature) of stars in different galaxies were compared with similar type of stars in the Milky Way, the stars become distributed according to a certain pattern, the so-called *Zero Age Main Sequence* (ZAMS) of stars, see Fig. 12.4. It is believed that most stars in their evolution follow the diagram beginning at the lower right side, along the main sequence into the red giant phase, then to the left and down, decreasing in size and temperature to end as blue or white dwarfs. *Hertzsprung-Russell* (*HR-*)*diagrams*, like the one in Figure 12.4, are valid for stars of about $0.7-70$ $\mathbf{M}_\odot$: from such diagrams conclusion can be drawn about the *size* (or *mass*) and *relative age* of the star, as it is assumed that stars of a given mass follow the same sequence as they age. The apparent luminosity, F, which we observe with our telescopes, is related to the *absolute luminosity*, L^*, i.e. the total *energy flux* in all directions from a star, by the relation

$$F = L^*/4\,\boldsymbol{\pi}\,d^2 \tag{12.1}$$

where d is the *distance* from the star. The historical classification of stars into *brightness classes* is now usually replaced by their relative (or apparent) *magnitude*, m^*, defined as

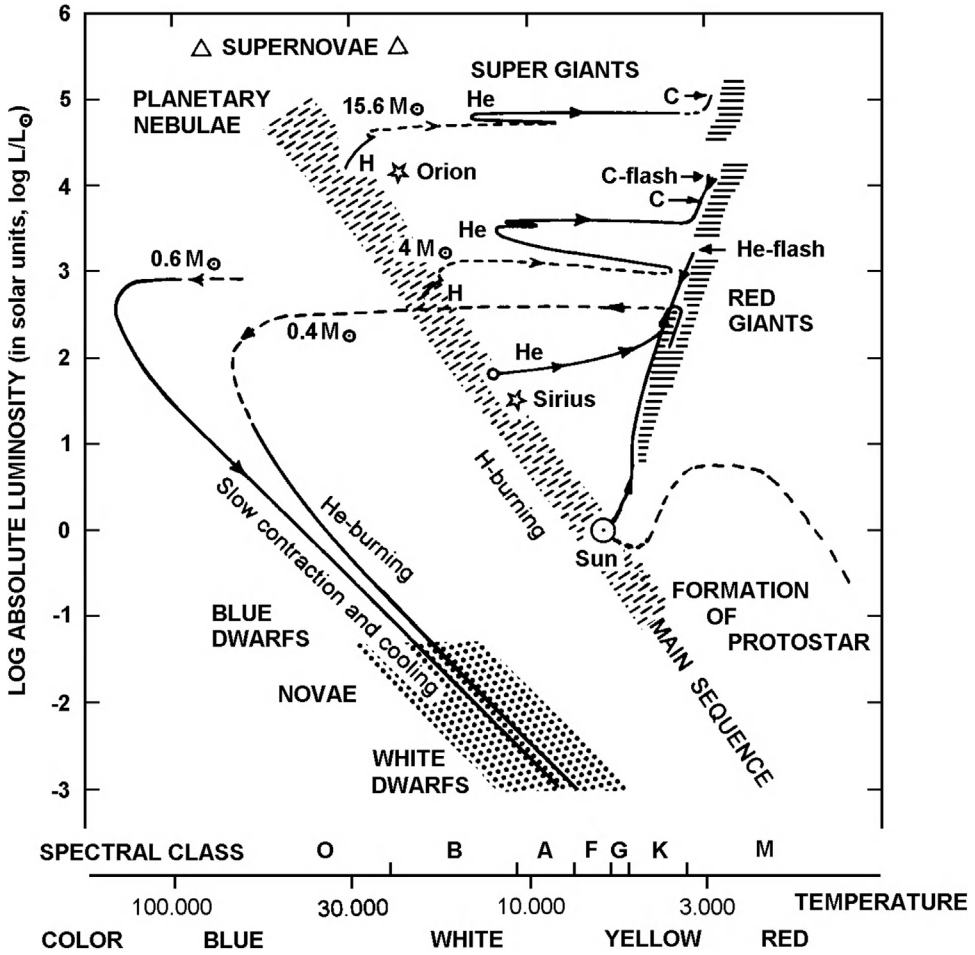


Figure 12.4 HR-diagram showing surface temperature, spectral class and color for stars born on the main sequence as a function of their absolute luminosity relative to the solar mass $M_\odot$.

$m^* = -2.5 \log(F/F_0)$, where F_0 is a reference flux density of light. Historically, the star Vega has been defined as having magnitude zero, but Vega is now known to vary its intensity somewhat with time and a better standard is needed.

Hubble discovered that all galaxies, except for those in the Virgo, show a *spectral red-shift* (i.e. increased distance between known frequency lines). This is assumed to be a Doppler effect due to that the objects move away from us (compare the lowering of the pitch from the horn of a train moving away from us). The red shift z is

$$z = (\lambda - \lambda_o)/\lambda_o = \mathbf{H}\, d/\mathbf{c} \tag{12.2}$$

where **H** is the *Hubble constant,* λ and λ_0 are the shifted and non-shifted wave length of a specific spectral line. For *velocities* $v \ll \mathbf{c}$, the relation becomes $z = v/\mathbf{c}$, hence

$$v_r = \mathbf{H}\, d \tag{12.3}$$

which is the common expression of the *Hubble law* and v_r is the *radial velocity.* If the red shift is plotted against the apparent magnitude of the brightest star in a large number of galaxies it is seen to increase with decreasing luminosity, which is interpreted so as that more distant (faintest) galaxies move away faster from us than the closer ones. Except for the galaxies in the Local Group, all galaxies recede from us with velocities up to 20 000 km/s; hence it is concluded that *the universe expands*. This conclusion and the underlying calculations was first published in French in a small Belgian scientific journal by a catholic priest − George Lemaître. At a later date Edvin Hubble draw the same conclusion from the same type of calculations using the same data and published a paper in English − and became famous.

12.1.5. Determination of ages

If the universe in general is expanding, the galaxies were once much closer to each other. If the rate of expansion had been unchanged, the inverse of the Hubble constant, $\mathbf{H}^{-1}$, would represent the *age of the universe.*

In (12.3) v_r is the radial velocity of a galaxy at distance d from us. But velocity is just distance divided by time; i.e. $v_r = d/t_o$, where t_o is the time the movement (expansion) has gone on, assuming a constant speed. Thus,

$$d/t_o = \mathbf{H}\, d \tag{12.4}$$

or

$$t_o = 1/\mathbf{H} \tag{12.5}$$

The numerical value of t_o is only an approximate estimate of the age of the universe, because we have all reasons to believe that the movement first slowed down due to gravitational pull and then began to accelerate again. According to the best present estimate a **H**-value of 70.4 ± 1.4 km s^{-1} pc^{-1}, after some corrections, corresponds to an age of 13.7 ± 0.13 Gy (giga years, 10^9 years, or *eons*). Cosmologists also give the age in the "scale factor" $(1 + z)$-values, i.e. red-shift values; e.g. we would observe a z-value of 10 for an object about one billion years old from the formation of the universe.

The *lifetime of a solar system* can be estimated from its composition and energy production. The average surface temperature of our sun is 5780 K while that of its center is about 1.5×10^7 K. The energy production rate is 3.76×10^{26} J s^{-1} $(= L)$ and is assumed to have been relatively constant since the formation of the sun about 5 eons ago. The energy production is so immense that gravitation alone cannot account for it, leading to the conclusion that the main solar energy source is not gravitational but nuclear. This is strongly supported by the cosmic abundance of the elements and measurements of the neutrino flux from the sun.

The high temperature allows hydrogen to fuse into helium by the formal reaction

$$4^1\mathrm{H} \rightarrow {}^4\mathrm{He} + \left|2\beta^+ + 2\mathrm{e}^-\right| + 2\nu_\mathrm{e} \quad Q = 24.7 + 2 \times 1.02 = 26.7\ \mathrm{MeV} \qquad (12.6)$$

(neutrinos take away 0.06 MeV). The particles within || annihilate and disappear. H-burning is assumed to be the main source of nuclear energy in a star of mass m. In 5 eons $\sim 9 \times 10^{28}$ kg ^{1}H was consumed, which is ~5% of the suns current content of ^{1}H. It is possible to calculate the time it takes to consume all available hydrogen; this is referred to as *the nuclear time scale*. From (12.6) one finds that about 0.70% of the hydrogen mass is converted into energy. As only about 10% of the hydrogen can be consumed before other more rapid evolutionary mechanisms set in (see below), the *nuclear lifetime*, t_n, is

$$t_n = 0.0070 \times 0.10 \times m\,\mathbf{c}^2/L \qquad (12.7)$$

which, for our sun ($m = \mathbf{M}_\odot$), gives a value of $\sim 10^{10}$ years, i.e. 10 eons.

The *age of matter* can be determined from its content of radioactive (and sometimes also stable) isotopes, as described in Chapter 13. From analyzes of lunar and meteoritic samples the age of the solar system has been calculated to be about 4.7 eons. The spectra of the light from some stars in our galaxy show the existence of relatively short lived elements like technetium (longest lived isotope has a half-life of 4.1×10^6 y). Because these elements cannot be formed in the main sequence stars, as we shall see later, they must have been accumulated "recently" from interstellar matter, indicating that these stars are rather young. From HR-diagrams, stars in our galaxy appears to have ages from <0.1 to >6 eons. Thus the evidence is that the formation of stars has gone on for a long period of time and is still occurring.

12.1.6. Elemental composition

Figure 12.5 shows the *relative abundance* of the elements of the universe and of the earth. The abundances are approximate, as a consequence of the difficulties in their assessment and limitations of experimental techniques. The abundances in the universe (based on spectral measurements on stars and interstellar matter) are used as a refinement of data obtained for the solar system. Stellar light is divided in spectral classes depending on the surface temperature of the star, see Fig. 12.4. The various classes (Harvard Spectral Classification) show lines of the elements as listed below in approximately decreasing intensity:

O	He^+, N^{2+} and Si^{3+}; weak H lines
B	He, $Si^{+,2+}$, O^+, Mg; more H than in Class O
A	Strong H; Mg^+, Si^+, Fe^+, Ti^+, Ca^+ and some other metals
F	H weaker than Class A; also Ca^+, Fe^+, Cr^+ and other metals
G	Lines of ionized metals; less H; also CH and hydrocarbons
K	Lines of neutral metals; also CH present
M	Lines of neutral metals; also molecular bands from C_2, CN, CH, TiO, ZrO, etc

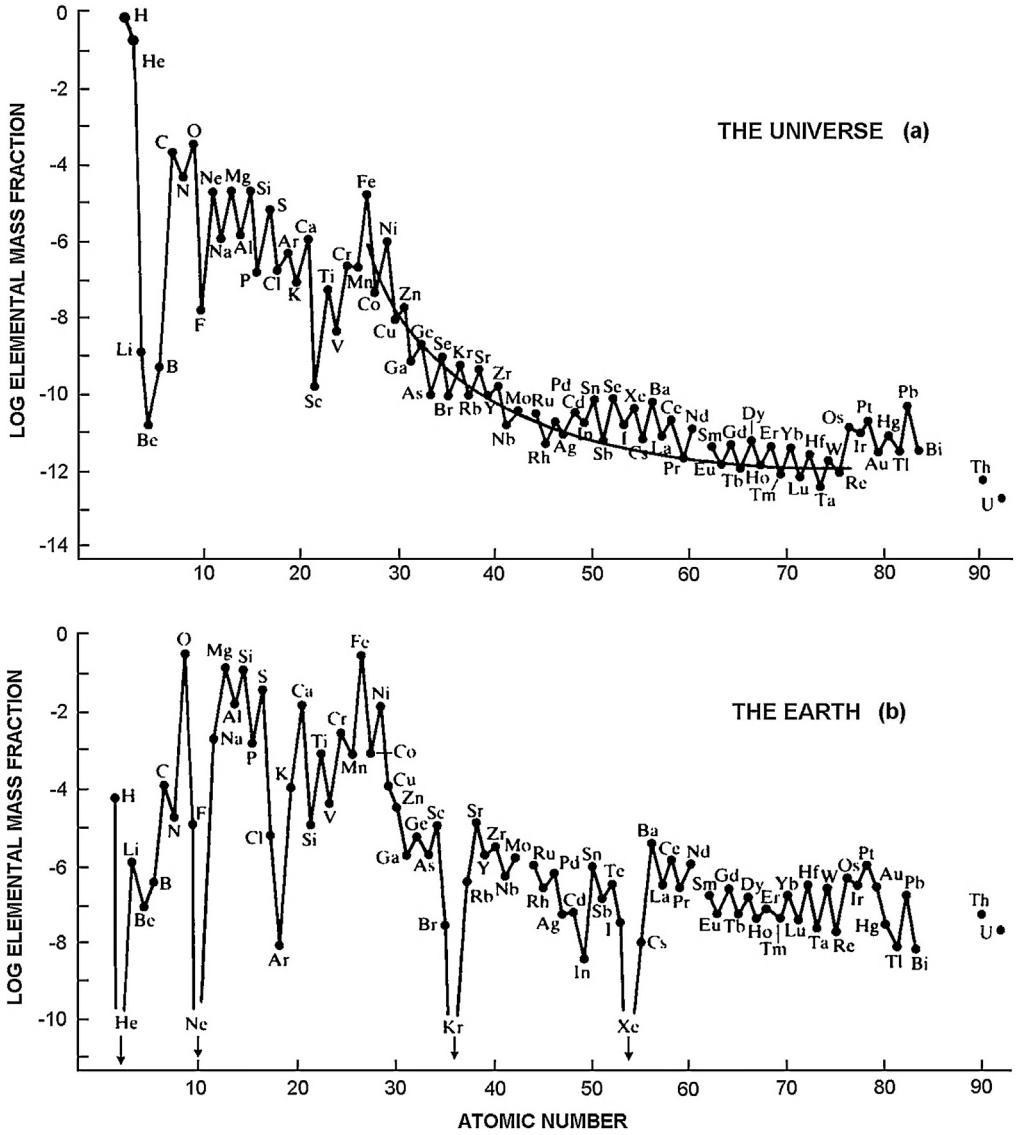


Figure 12.5 Relative abundances of the elements in the universe and the earth. *(From Cox.)*

Some stars are labelled chemically peculiar (CP-stars) because the spectrum of their light is different from most stars in the main sequence, e.g. they have very strong lines of rare earths or heavy elements. Strong lines are assumed to reflect that these elements have been moved up from the core by heavy turbulence. The Hubble Space Telescope (HST), which was placed in orbit around the earth in 1990, has allowed detailed spectroscopic analysis of CP-stars. For example in X-Lupi, high concentrations of heavy metals like

Hg, Ru and Pt have been discovered; and 100% of the mercury is ^{204}Hg, while the solar abundance of that isotope is only 7%.

The abundance data in Figure 12.5 show some regularities:

(i) H and He are the dominating elements in the universe (>99 %),

(ii) the abundances show a rapid, exponential decrease with increasing Z,

(iii) there is a pronounced abundance peak $Z = 26$ (Fe), and minor peaks at 6 and 8 (C and O),

(iv) elements of even atomic number are more abundant than their neighbours of odd atomic number.

Further, a study of isotopic abundances shows:

(a) for the lightest elements, isotopes of masses which are multiples of 4 show the highest abundances, indicating their formation through fusion of He-atoms;

(b) for isobars lighter than $A \approx 70$ (corresponding to $Z \approx 28$) the most proton rich (or most neutron deficient) isobar is the most abundant, while the opposite goes for isobars with $A > 70$, indicating two different modes of synthesis for elements lighter than and heavier than $Z \approx 28$;

(c) elements that have N and Z values close to the magic numbers (Ch. 6) are more abundant then neighbouring ones, further supporting the idea of elemental formation through nuclear processes.

From these and additional observations Burbidge, Burbidge, Fowler, and Hoyle ("B^{2}FH") and others have developed a model for elemental formation in the universe, which is the basis for much of the subsequent discussion.

12.1.7. Microwave radiation and the big bang

In 1965 it was discovered that low energy microwave radiation (at 7.35 cm uncorrected) reaches us from all directions in space (about 400 photons cm^{-3}). This is referred to as the *cosmic background radiation* whose wavelength corresponds to radiation from a black body of temperature 2.725 K (about 0.0003 eV). Thermodynamic calculations show that this is the temperature reached after adiabatic expansion of a very hot cloud for some 10 billion years. The existence of such a background radiation was predicted by Gamow decades before in a cosmological hypothesis referred to as the *Big Bang model*. Recent measurements have shown that the intensity of this radiation varies slightly ("ripples") in different directions in space.

The Big Bang hypothesis requires an instantaneous beginning of our universe at a point to which all energy (in the Planck-Einstein matter-radiation-energy concept, Ch 10) is concentrated. Ordinary nuclear reactions cannot model this beginning and we must turn to particle physics for information (Ch 2).

In the following we describe the formation of the universe and the elements according to the so-called *Standard Model of Stellar Evolution*, which is based on the models originally developed by Bethe and Weizsäcker in the 1930's for the reactions in the sun,

and the Big Bang hypothesis for the formation of the universe as originally suggested in 1948 by Gamow, Alpher and Herman, and later developed by the B^2FH-group, Weinberg and others.

12.2. IN THE BEGINNING OF TIME

Around "time zero" the Universe consisted of an immensely dense, hot sphere of photons, quarks and leptons, and their antiparticles, in thermal equilibrium, particles being created by photons and photons created by annihilation of particles. The temperature must have been $\geq 10^{13}$ K, but no light was emitted, because of the enormous gravitational force pulled the photons back. The system was supposed to be in a unique state with no repulsion forces. However, just as a bottle of supercritical (overheated) water can explode by a phase transition, so did the Universe, space came into being and time began. The Universe expanded violently in all directions, and as age and size grew, density and temperature fell.

A one hundreds of a second later all the quarks were gone, and the Universe consisted of an approximately equal number of electrons, positrons, neutrinos and photons, and a small amount of protons and neutrons; the ratio of protons to photons is assumed to have been about 10^{-9}. The temperature was about 10^{11} K and the density so high, about $4 \times 10^6\,\mathrm{kg\,m^{-3}}$, that even the unreactive neutrinos were hindered to escape.

The conditions can be partly understood by considering the relations

$$E\,(\mathrm{MeV}) = m\,\mathbf{c}^2 = 931.5\;\Delta M\,(\mathrm{u}) \qquad (4.6 \text{ and } 12.8)$$

which gives the energy required to create a particle of mass ΔM, and

$$E\,(\mathrm{MeV}) = \mathbf{k}\;T = 8.61 \times 10^{-11}\,T \qquad (3.53 \text{ and } 12.9)$$

which gives the average kinetic energy of a particle at temperature T (K, Kelvin). As the photon energy of E (eV) corresponds to the wavelength λ (m) according to (3.52)

$$E\,(\mathrm{eV}) = \boldsymbol{h}\,\nu = \boldsymbol{h}\,\mathbf{c}/\lambda = 1.240 \times 10^{-6}/\lambda\,\left(\mathrm{m}\right) \qquad (12.10)$$

one can estimate that the creation of a proton or a neutron (rest mass 940 MeV) out of radiation requires a temperature of 1.1×10^{13} K, corresponding to a photon wave length of about 10^{-15} m, i.e. the size of a nucleon. At these temperatures nucleons are formed out of radiation, but are also disrupted by photons, leading to an equilibrium with about an equal number of protons and neutrons. At temperatures below the threshold formation energy, no nucleons are formed. However, it should be remembered that particles and radiation are distributed over a range of energies, according to the Boltzmann (cf. Fig. 3.4) and Planck distribution laws. Thus some formation (and disruption) of nucleons occurs even at lower temperatures.